

Advanced Machine Learning Project:
Healthcare Workers Mental Health Prediction

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Executive Summary

This project develops a machine learning model to predict mental health risk among healthcare workers using workplace and personal factors. Using a dataset of 5,000 healthcare employees, we implemented and compared Logistic Regression and Random Forest algorithms to identify workers at high risk for mental health issues.

Key Findings:

- Logistic Regression achieved 94.1% accuracy in predicting mental health risk
- 65.9% of healthcare workers in the dataset are classified as high-risk
- Stress level (correlation: 0.737) and workplace factors (correlation: -0.552) are the strongest predictors
- General Practice and ICU departments show the highest risk rates (83.6% and 83.2% respectively)

The model can be used by healthcare organizations to identify at-risk employees and implement targeted mental health interventions.

Introduction

Healthcare workers face unique challenges that significantly impact their mental health, including high-stress environments, long working hours, emotional demands, and life-or-death decision-making responsibilities. The COVID-19 pandemic has further exacerbated these challenges, leading to increased rates of burnout, anxiety, and depression among healthcare professionals.

Mental health issues in healthcare settings not only affect individual well-being but also impact patient care quality, workplace safety, and organizational productivity. Early identification of at-risk employees can enable proactive interventions, reduce turnover rates, and improve overall healthcare delivery.

This project leverages machine learning techniques to develop a predictive model that can identify healthcare workers at high risk for mental health issues based on workplace factors, job characteristics, and personal indicators.

Problem Statement

Research Question

Can we accurately predict mental health risk among healthcare workers using machine learning models trained on workplace factors, job satisfaction, stress levels, and organizational characteristics?

Objectives

- i. Develop a binary classification model to predict mental health risk (High Risk vs. Low Risk)
- ii. Identify the most significant predictors of mental health risk in healthcare settings
- iii. Analyze risk patterns across different employee types, departments, and workplace factors
- iv. Provide actionable recommendations for healthcare organizations to support employee mental health

Success Metrics

- Achieve classification accuracy $> 90\%$
- Identify key risk factors with statistical significance
- Provide interpretable results for organizational decision-making

Literature Review

Mental health challenges among healthcare workers have been extensively documented in recent literature. Key findings from previous research include:

Stress and Burnout Factors: High workload and time pressure are primary contributors to healthcare worker stress (Maslach & Leiter, 2016)

Poor work-life balance significantly correlates with mental health issues (Shanafelt & Noseworthy, 2017)

Job satisfaction serves as a protective factor against burnout (West et al., 2018)

Organizational Factors: Access to Employee Assistance Programs (EAPs) can reduce mental health risks (Joyce et al., 2019)

Department-specific risk variations exist, with ICU and emergency departments showing higher rates (Prasad et al., 2021)

Workplace factors such as safety concerns and work environment quality impact mental health outcomes (Smith et al., 2020)

Machine Learning Applications: Previous studies have used logistic regression and ensemble methods for healthcare worker burnout prediction (Chen et al., 2021)

Feature importance analysis has identified stress level and job satisfaction as primary predictors (Kumar et al., 2022)

Dataset Description

Data Source

The dataset was obtained from Kaggle: "Healthcare Workforce Mental Health Dataset" (Rivalytics, 2024). The dataset contains comprehensive information about healthcare workers' workplace conditions, personal characteristics, and mental health indicators.

Dataset Characteristics

- Size: 5,000 healthcare employees
- Features: 10 variables (7 categorical, 3 numerical)
- Time Period: Current healthcare workforce data
- Data Quality: No missing values, clean dataset

Variables Description

- Employee Types: Diverse representation across nursing roles, technicians, and support staff
- Departments: Balanced distribution across hospital departments
- Stress Levels: Mean: 7.2, indicating generally high stress levels
- Job Satisfaction: Mean: 2.8, indicating moderate to low satisfaction

VARIABLE	TYPE	DESCRIPTION	VALUES/RANGE
Employee ID	Categorical	Unique identifier	HCP-00001 to HCP-05000
Employee Type	Categorical	Job role category	6 types (Nurse, Technician, etc.)
Department	Categorical	Work department	15 departments

Workplace Factor	Categorical	Primary workplace stressor	4 categories
Stress Level	Numerical	Self-reported stress (1-10 scale)	Range: 4-9
Burnout Frequency	Categorical	Frequency of burnout episodes	Never, Occasionally, Often
Job Satisfaction	Numerical	Job satisfaction rating (1-5 scale)	Range: 1-5
Access to EAPs	Categorical	Employee Assistance Program access	Yes/No
Mental Health Absences	Numerical	Days absent due to mental health	Range varies
Turnover Intention	Categorical	Intent to leave job	Yes/No

Methodology

Target Variable Definition

I created a binary target variable `mental_health_risk` using a rule-based approach:

- High Risk (1): $\text{Stress Level} \geq 7$ AND ($\text{Job Satisfaction} \leq 2$ OR $\text{Turnover Intention} = \text{'Yes'}$)
- Low Risk (0): All other cases

This definition captures employees experiencing both high stress and either low job satisfaction or intent to leave, indicating elevated mental health risk.

Data Preprocessing Steps

- Data Cleaning: Verified no missing values, confirmed data types
- Categorical Encoding:
- Used `pd.factorize()` for Employee Type and Department
- Manual mapping for ordinal variables (Burnout Frequency, EAP Access)
- Feature Scaling: Applied `StandardScaler` to normalize feature ranges
- Train-Test Split: 70% training, 30% testing with stratified sampling

Machine Learning Algorithms

1. Logistic Regression

- Linear classification algorithm + Interpretable coefficients
- Good baseline model for binary classification
- Assumptions: Linear relationship between features and log-odds

2. Random Forest

- Ensemble method using multiple decision trees
- Handles non-linear relationships
- Provides feature importance rankings
- Robust to outliers and overfitting

3. XGBoost (Extreme Gradient Boosting)

- Advanced gradient boosting algorithm
- Often achieves state-of-the-art performance
- Handles complex feature interactions
- Built-in regularization to prevent overfitting

Model Evaluation Metrics

- Accuracy: Overall correct predictions
- Precision: $\text{True positives} / (\text{True positives} + \text{False positives})$
- Recall: $\text{True positives} / (\text{True positives} + \text{False negatives})$
- F1-Score: Harmonic mean of precision and recall
- Confusion Matrix: Detailed classification results

Implementation

Technical Environment

- Programming Language: Python 3.x
- Development Environment: Jupyter Notebook

Key Libraries:

- pandas: Data manipulation and analysis
- numpy: Numerical computations
- scikit-learn: Machine learning algorithms

- matplotlib/seaborn: Data visualization

Code Structure

Main implementation components:

1. Data Loading and Exploration
2. Data Preprocessing and Feature Engineering
3. Exploratory Data Analysis
4. Model Training and Evaluation
5. Results Analysis and Visualization

Detailed Performance Analysis:

Logistic Regression achieved the best overall performance with 94.1% accuracy.

1. Logistic Regression Results:

- Class 0 (Low Risk): Precision: 92%, Recall: 91%
- Class 1 (High Risk): Precision: 95%, Recall: 96%
- Overall: Strong performance across both classes; slight bias toward detecting high-risk cases

2. Feature Importance Analysis

- Correlation with Mental Health Risk:
- Stress Level: 0.737 (Strong positive correlation)
- Workplace Factor: -0.552 (Strong negative correlation)
- Job Satisfaction: -0.533 (Strong negative correlation)
- Mental Health Absences: 0.351 (Moderate positive correlation)
- Burnout Frequency: 0.322 (Moderate positive correlation)

Key Insights:

- Stress level is the strongest predictor of mental health risk
- Poor workplace factors (safety concerns, heavy workload) increase risk
- Higher job satisfaction provides protective effect
- Employee type and department show minimal correlation

Risk Distribution Analysis

Overall Risk Statistics:

- High-Risk Employees: 3,293 out of 5,000 (65.9%)
- Average Stress Level (High-Risk): 8.1/10
- Average Job Satisfaction (High-Risk): 1.8/5

Department Risk Analysis:

- General Practice: 83.6% high-risk
- ICU: 83.2% high-risk
- General Medicine: 75.1% high-risk
- Emergency Department: 72.3% high-risk
- Pediatrics: 68.9% high-risk

Workplace Factor Impact:

- Heavy Workload: Highest risk association
- Safety Concerns: Second highest risk factor
- Poor Work Environment: Moderate risk increase
- Work-Life Imbalance: Significant risk contributor

Discussion

Model Performance Interpretation

The Logistic Regression model's 94.1% accuracy indicates strong predictive capability for identifying at-risk healthcare workers. This performance level is suitable for practical implementation in healthcare organizations.

Strengths of the Logistic Regression Model:

- High interpretability of results
- Strong performance across both risk classes
- Computationally efficient
- Provides probability scores for risk assessment

Model Reliability Factors:

- Balanced precision and recall indicate robust classification
- No significant overfitting observed (training vs. test performance similar)
- Feature correlations align with domain knowledge

Feature Analysis Insights

- *Primary Risk Factors:* Stress Level (0.737 correlation): The dominant predictor, confirming that self-reported stress is a reliable indicator of mental health risk

- *Workplace Factors*: Negative correlation (-0.552) indicates that certain workplace conditions significantly impact mental health
- *Job Satisfaction*: Strong protective factor (-0.533), suggesting that improving job satisfaction can reduce risk

Surprising Findings:

- Employee type and department show minimal direct correlation with risk
- Access to EAPs shows very weak correlation (-0.007), suggesting that availability alone may not be sufficient

Departmental Risk Patterns

The high-risk percentages in General Practice (83.6%) and ICU (83.2%) departments align with literature findings about high-stress medical environments. These departments typically involve:

- High patient acuity and mortality rates
- Time-critical decision making
- Heavy emotional burden
- Extended work hours

Implications for Healthcare Organizations

Immediate Actions:

- (i) Focus screening efforts on employees with stress levels ≥ 7
- (ii) Prioritize interventions in General Practice and ICU departments
- (iii) Address workplace factors causing stress (workload, safety, environment)

Long-term Strategies:

- (i) Implement job satisfaction improvement programs
- (ii) Review EAP effectiveness and utilization
- (iii) Develop department-specific mental health support programs

Conclusions and Recommendations

- **Model Effectiveness:** Machine learning can accurately predict mental health risk in healthcare workers (94.1% accuracy)
- **Key Risk Factors:** Stress level, workplace factors, and job satisfaction are the most important predictors
- **High-Risk Prevalence:** Nearly two-thirds of healthcare workers in the dataset are at high risk for mental health issues
- **Department Variations:** Certain departments (General Practice, ICU) require targeted interventions

Limitations and Future Work

Data Limitations:

- a. **Cross-sectional Data:** Snapshot at single time point, cannot capture temporal changes
- b. **Self-reported Measures:** Potential bias in stress level and job satisfaction responses
- c. **Limited Demographics:** Missing age, gender, and experience level data
- d. **Binary Classification:** Simplified risk categorization may miss nuanced risk levels

Methodological Limitations:

- a. **Target Variable Definition:** Rule-based approach may not capture all aspects of health risk
- b. **Feature Selection:** Limited to available dataset features, missing potentially important factors
- c. **External Validity:** Results may not generalize to other healthcare systems or regions

Future Research Directions

- **Longitudinal Studies:** Track employees over time to understand risk progression
- **Additional Features:** Include demographic data, work experience, and personal factors
- **Clinical Validation:** Correlate predictions with clinical mental health assessments
- **Multi-site Studies:** Validate model across different healthcare organizations
- **Deep Learning:** Explore neural networks for complex pattern recognition
- **Ensemble Methods:** Combine multiple algorithms for improved accuracy
- **Time Series Analysis:** Predict risk trajectory over time
- **Unsupervised Learning:** Discover hidden patterns in mental health data
- **Intervention Effectiveness:** Measure impact of model-guided interventions
- **Cost-Benefit Analysis:** Quantify economic benefits of early risk identification
- **User Acceptance:** Study healthcare worker and manager adoption of predictive tools
- **Model can handle larger datasets with minimal modification**
- **Cloud deployment for real-time risk assessment**
- **Integration with existing HR information systems**

- Framework adaptable to different healthcare settings
- Customizable risk thresholds based on organizational priorities
- Training programs for implementation teams

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Dataset Source

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